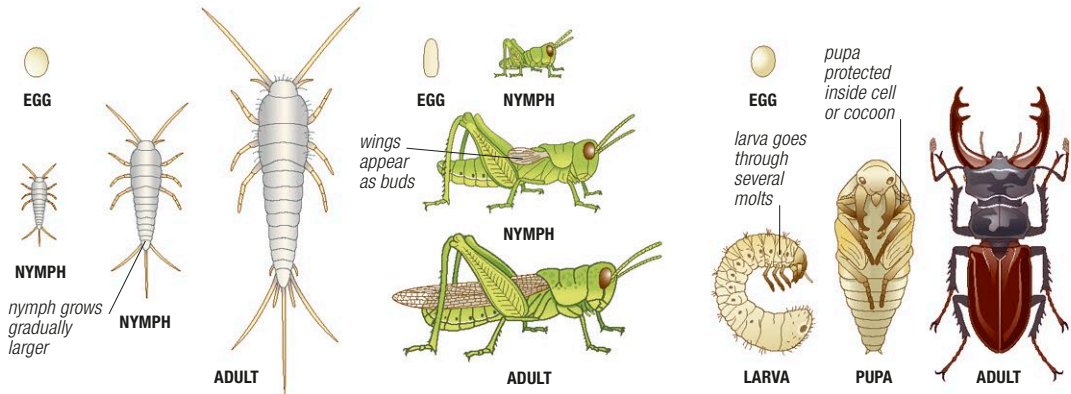


Life cycles

Insects begin life as an egg. However, once a young insect hatches, it may follow one of several routes to adulthood (see right), varying from a simple series of molts to a complete physical transformation of its body parts. The length of each stage varies greatly between species. For example, a few species of North American cicadas take 17 years to reach adulthood, while some fruit flies may take less than 2 weeks to mature. Once insects reach sexual maturity, they begin to mate. Courtship displays may involve the production of sexual odors, sounds, and even light displays. In almost all insects, fertilization is internal (see pp.30–1). The fertilized eggs are usually laid close to a source of food.

However, in parasitic species, they may be laid either on or inside the body of a host animal.



AMETABOLOUS DEVELOPMENT

In some wingless insects, there is little difference in form between young and adults. The nymph develops by shedding its exoskeleton, growing larger after each molt.

INCOMPLETE METAMORPHOSIS

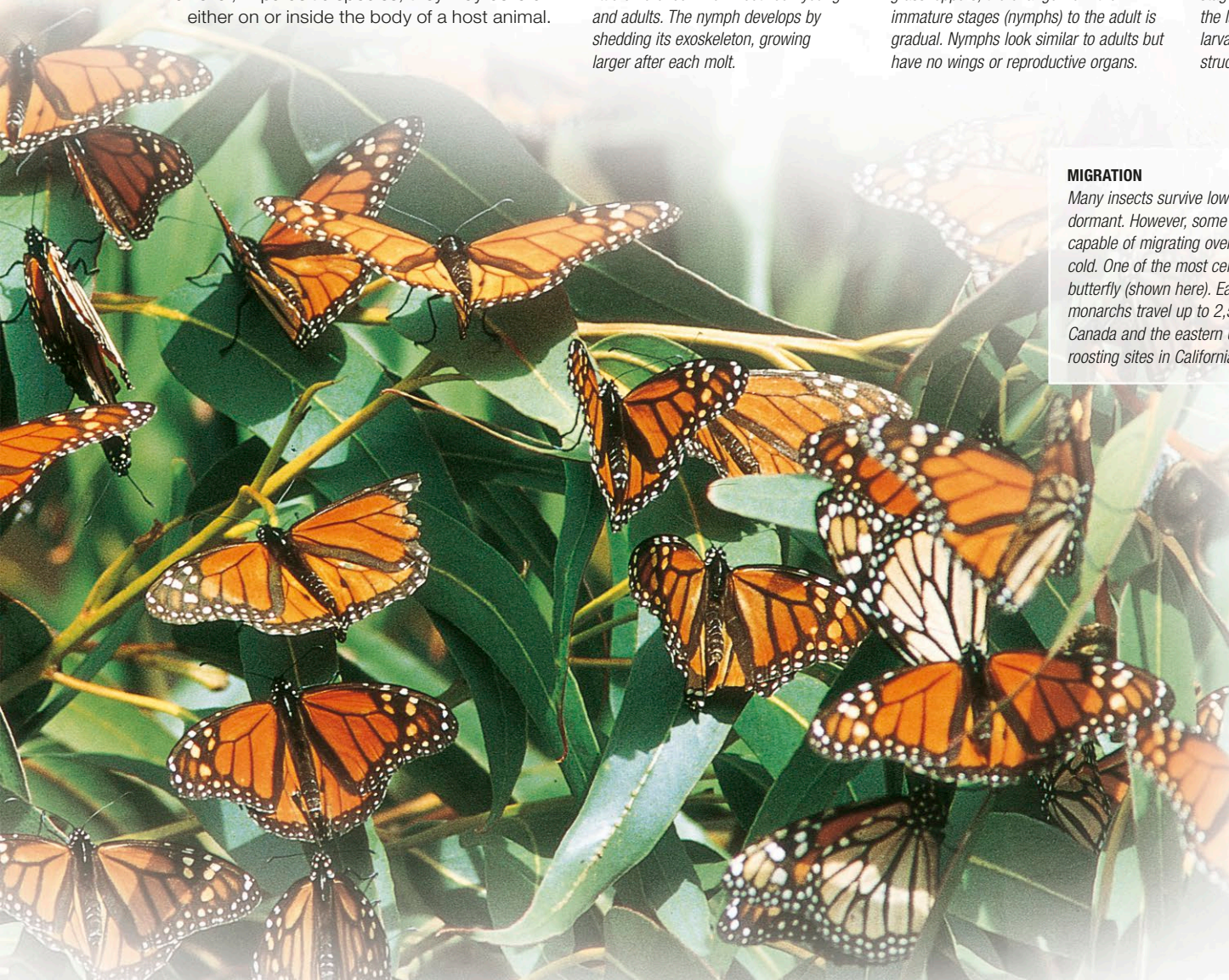
In some winged insects, such as grasshoppers, the change from the immature stages (nymphs) to the adult is gradual. Nymphs look similar to adults but have no wings or reproductive organs.

COMPLETE METAMORPHOSIS

In other winged insects, the immature stages (larvae) molt several times. During the last molt, the larvae pupate and the larval tissues are transformed into adult structures.

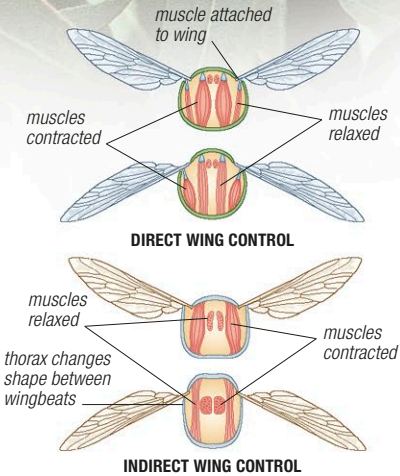
MIGRATION

Many insects survive low temperatures by becoming dormant. However, some of the larger winged insects are capable of migrating over great distances to avoid winter cold. One of the most celebrated examples is the monarch butterfly (shown here). Each year, tens of millions of monarchs travel up to 2,500 miles (4,000 km) from Canada and the eastern United States to winter in roosting sites in California and Mexico.



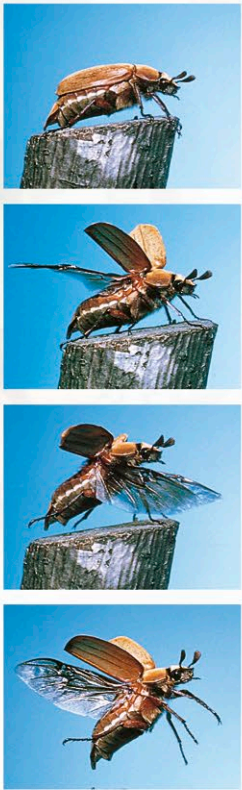
Flight and wings

Insects were the first animals to evolve the power of flight, a development that made it easier for them to evade predators and to find food and mates efficiently. An insect's wings, which are made of cuticle, are hinged to the thorax. Most insects have 2 pairs of wings. These are usually hooked together, but in some species, such as dragonflies, they beat alternately, which improves maneuverability. True flies are the only insects that have one pair of wings; they are also the most agile fliers, capable of flying backward, sideways, and even upside down. Almost all flying insects can fold their wings, which means that when they are not flying they can squeeze into small spaces, such as crevices in bark or gaps under stones.



DIRECT AND INDIRECT WING CONTROL

Winged insects use various mechanisms to control their wings. Some control them directly, using muscles attached to the wing bases to produce the upstroke. Other insects move their wings up and down by changing the shape of the thorax. Insects that use this indirect control mechanism usually beat their wings more quickly (100–400 times per second) than those that use direct control (usually about 50 times per second).



TAKING OFF

Beetles protect their delicate hindwings under tough, caselike forewings (elytra). It takes a little time for a beetle to lift its elytra and unfold the hindwings before taking off. If the air is cold, many insects vibrate their flight muscles to warm them up.